SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – SCENARIO BASED (ANSWERED)

Course Code:	DSA0313	**Course Title:**	Natural Language Processing
CO Assessed:	CO2 – Manipulation	**Assessment:**	Assessment Tool 2 – SCENARIO BASED
Weightage:	35% of CIA		
CO5 Statement:	Design inflectional, derivational, finite-state parsing, stemming algorithms and analyse morphological methods. (BL-3)		
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Section / Batch: _____ Date: 04-11-2026

Code of Conduct

I G V Durga Charan (192324100) (Name / Reg. No.) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: G V Durga Charan

Instructions

- Read all questions carefully before attempting the answers.
- Answer all five questions. Each question carries equal marks.
- Show morphological decomposition, rules, classifications, and justifications wherever applicable.
- Duration: 30 minutes.

Annexure A – SCENARIO BASED QUESTIONS WITH ANSWERS

1. Morphological processing of analyzing, analysis, and analytical

Scenario: A large-scale search engine indexing system processes user queries containing “ analyzing” , “ analysis” , and “ analytical” . The system must group semantically related terms under a unified representation. Design a morphological processing approach to decompose each input word into root and affixes, distinguish inflectional and derivational forms, and normalize them for indexing.

Answer / Morphological analysis: The common lexical base is ANALYSE (often represented as “ analyze” in American English). “ Analyzing” is an inflectional form of the verb; “ analysis” and “ analytical” are derivationally related forms. A simple suffix-stripping rule alone cannot safely derive all three forms, so an explicit exception/lexicon mapping is used.

Python program:

```
words = ["analyzing", "analysis", "analytical"]

# Explicit lexical mappings handle spelling changes and irregular derivations.
morphology = {
    "analyzing": {
        "root": "analyze", "prefix": "", "suffix": "-ing",
        "type": "Inflectional", "normalized": "analyze"
    },
    "analysis": {
        "root": "analysis", "prefix": "", "suffix": "",
        "type": "Derivational/related noun (lexical form)",
        "normalized": "analyze"
    },
```

```
    "analytical":{
        "root":"analysis","prefix":"","suffix":"-ical",
        "type":"Derivational(adjectiveformation)",
        "normalized":"analyze"
    }
}

print(f'{"Word":15}{"Root":12}{"Affix":10}{"Type":35}Normalized')
forwordinwords:
    item=morphology[word]
    affix=(item["prefix"]+item["suffix"])or"None"
    print(f'{word:15}{item["root"]:12}{affix:10}'
          f'{item["type"]:35}{item["normalized"]}')
```

Result and conclusion: Expected report:

- analyzing → analyze + -ing; inflectional present participle; normalized: analyze.
- analysis → lexical/related noun form; derivational relationship to analyze; normalized: analyze.
- analytical → analysis + -ical; derivational adjective; normalized: analyze.

The index stores the normalized key “ analyze” while retaining the original word for display.

2. Morphological parsing of disagree, agreement, and agreeable

Scenario: An AI-based sentiment analysis platform receives product reviews containing “ disagree” , “ agreement” , and “ agreeable” . Identify prefixes, suffixes, and base forms, and capture how derivational changes influence meaning.

Answer / Morphological analysis: The base is “ agree” . The prefix “ dis-” reverses or negates the meaning. The suffix “ -ment” forms a noun denoting an action, process, or result. The suffix “ -able” forms an adjective meaning capable of or suitable for agreeing.

Python program:

```
words=["disagree","agreement","agreeable"]

rules={
    "disagree":{
        "prefix":"dis-","root":"agree","suffix":"",
        "category":"Derivational(prefixation)",
        "meaning":"notagree;expressadifferentopinion"
    },
    "agreement":{
        "prefix":"","root":"agree","suffix":"-ment",
        "category":"Derivational(nounformation)",
        "meaning":"thestate,act,orresultofagreeing"
    },
    "agreeable":{
        "prefix":"","root":"agree","suffix":"-able",
        "category":"Derivational(adjectiveformation)",
        "meaning":"pleasantorwillingtoagree"
    }
}

print(f'{"Word":13}{"Prefix":8}{"Root":8}{"Suffix":9}'
      f'{"Type":32}Meaning')
forwordinwords:
    x=rules[word]
    print(f'{word:13}{x["prefix"]or"-":8}{x["root"]:8}'
          f'{x["suffix"]or"-":9}{x["category"]:32}{x["meaning"]}')
```

Result and conclusion: All three words share the base “ agree” , but their meanings and grammatical classes differ:

- disagree = dis- + agree (verb): opposite meaning.
- agreement = agree + -ment (noun): action/state/result.
- agreeable = agree + -able (adjective): having a pleasant or willing quality.

For sentiment analysis, preserve the derived meaning; do not treat every form as sentimentally identical.

3. Morphology-based normalization of govern, government, and governance

Scenario: A news aggregation system processes “ govern” , “ government” , and “ governance” . Standardize these variations for topic modeling and clustering by identifying the root, derivational suffixes, and derivational levels.

Answer / Morphological analysis: The shared root is “ govern” . “ Government” is formed with “ -ment” , producing a noun referring to the act, system, or body involved in governing. “ Governance” is formed with “ -ance” , producing a noun referring to the process or manner of governing. Both are derivational forms.

Python program:

```
words = ["govern","government","governance"]

analysis = {
    "govern": {
        "root": "govern","affix": "None",
        "level": "Level 0 – base verb",
        "normalized": "govern"
    },
    "government": {
        "root": "govern","affix": "-ment",
        "level": "Level 1 – verb to noun",
        "normalized": "govern"
    },
    "governance": {
        "root": "govern","affix": "-ance",
        "level": "Level 1 – verb to noun",
        "normalized": "govern"
    }
}

print(f'{"Word":13}{"Root":10}{"Affix":8}'
      f'{"Derivational level":27}Normalized')
for word in words:
    x = analysis[word]
    print(f'{word:13}{x["root"]:10}{x["affix"]:8}'
          f'{x["level"]:27}{x["normalized"]}')
```

Result and conclusion: Morphological breakdown:

- govern → govern; base verb; level 0.
- government → govern + -ment; derived noun; level 1.
- governance → govern + -ance; derived noun; level 1.

The common normalized representation is “ govern” . In an actual topic model, the original forms should also be retained because “ government” and “ governance” can express different concepts.

4. Morphological parsing of activate, activation, and reactivation

Scenario: A document classification system processes “ activate” , “ activation” , and “ reactivation” . Identify prefixes, roots, suffixes, derivational layers, word-class changes, and semantic shifts, then map the words to a common base representation.

Answer / Morphological analysis: The root is “ act” . The word “ activate” contains the derivational suffix “ -ivate” and functions as a verb meaning to make active. “ Activation” is formed from “ activate” with “ -ion” , producing a noun denoting the process. “ Reactivation” adds the prefix “ re-” to “ activation” , meaning activation again.

Python program:

```
words = ["activate","activation","reactivation"]

analysis = {
    "activate": {
        "prefix": "-", "root": "act", "suffix": "-ivate",
        "sequence": "act → activate",
        "class": "Verb","meaning": "to make active"
    },
    "activation": {
        "prefix": "-", "root": "act", "suffix": "-ivate + -ion",
        "sequence": "act → activate → activation",
        "class": "Noun","meaning": "the process of making active"
```

```
    },
    "reactivation":{
        "prefix":"re-","root":"act","suffix":"-ivate+-ion",
        "sequence":"act → activate → activation → reactivation",
        "class":"Noun","meaning":"the process of activating again"
    }
}

print(f'{"Word":15}{"Prefix":8}{"Root":7}{"Suffix":18}'
      f'{"Class":8}Meaning')
for word in words:
    x = analysis[word]
    print(f'{word:15}{x["prefix"]:8}{x["root"]:7}'
          f'{x["suffix"]:18}{x["class"]:8}{x["meaning"]}')
    print("   Derivation:",x["sequence"])
```

Result and conclusion: Derivational sequence and normalization:

- activate → act + -ivate; verb; normalized base: activate.
- activation → activate + -ion; noun; normalized base: activate.
- reactivation → re- + activation; noun; " again" is added to the meaning; normalized base: activate.

The root " act" is common to all three, while " activate" is a useful lexical normalization target for grouping documents about the action.

5. Inflectional normalization of create, creates, and creating

Scenario: A search optimization module processes “ create” , “ creates” , and “ creating” . Identify inflectional suffixes, map the forms to a common base without changing lexical meaning, and classify each grammatical transformation.

Answer / Morphological analysis: All three forms have the lemma “ create” . “ Create” is the base form. “ Creates” adds “ -s” for third-person singular present tense. “ Creating” adds “ -ing” to form the present participle/gerund. These are inflectional changes and do not create a new lexical word.

Python program:

```
words = ["create", "creates", "creating"]

def analyze(word):
    if word == "create":
        return "", "base form", "create"
    if word == "creates":
        return "-s", "third-person singular present", "create"
    if word == "creating":
        return "-ing", "present participle / gerund", "create"
    return "unknown", "unknown", word

print(f'{"Word":12}{"Suffix":8}{"Grammatical category":34}Root')
for word in words:
    suffix, category, root = analyze(word)
    print(f'{word:12}{suffix or "None":8}{category:34}{root}')
```

Result and conclusion: Normalization:

- create → no suffix; base form; lemma: create.
- creates → -s; third-person singular present; lemma: create.
- creating → -ing; present participle/gerund; lemma: create.

All variants are indexed under “ create” . The suffix is removed only after checking the grammatical pattern, preserving the same lexical meaning.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Clear and complete understanding	Good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand
Identification of Key Issues	20%	Identifies all relevant issues	Identifies most issues with minor omissions	Identifies limited issues	Unable to identify key issues
Application of Concepts	25%	Applies concepts effectively	Applies concepts with minor errors	Limited or partially correct application	Unable to apply concepts
Decision Making	20%	Logical decisions with strong rationale	Reasonable decisions with some justification	Weak decisions with limited justification	No clear decisions or rationale
Clarity of Response	15%	Clear, structured, logical response	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Assessment Record

Q. No.	Understanding	Key Issues	Application	Decision Making	Clarity	Sub. Total	Total %	Total %
1								
2								
3								
4								

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____